THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In science, work is done when: 1

(a) a person holds a heavy load at rest
(b) a force acts on a body and the body is displaced in the direction of the force
(c) a force acts but there is no displacement
(d) a person thinks hard

2. The work-energy theorem states that the work done on a body is equal to the change in its: 1

(a) momentum **(b)** mass **(c)** kinetic energy **(d)** potential energy

3. A simple machine is a device that: 1

(a) needs no effort
(b) creates energy
(c) does more work than the input
(d) makes work easier by changing the magnitude or direction of force

4. The SI unit of work is the: 1

(a) pascal **(b)** newton **(c)** watt **(d)** joule

5. The law of conservation of energy states that energy: 1

(a) can neither be created nor destroyed but only changed from one form to another
(b) is always lost
(c) can be created
(d) can be destroyed

6. Mechanical advantage = 1

(a) effort / load **(b)** load / effort **(c)** load - effort **(d)** load x effort

7. A girl carrying a box walks horizontally. The work done by the force holding the box is: 1

(a) negative **(b)** positive **(c)** zero **(d)** maximum

8. Power is the: 1

(a) force **(b)** work done
(c) rate of doing work **(d)** energy

9. A lever is a rigid bar that rotates about a fixed point called the: 1

(a) axle **(b)** effort **(c)** load **(d)** fulcrum

10. A force of 10 N moves a body through 5 m in the direction of the force. The work done is: 1

(a) 5 J **(b)** 15 J **(c)** 50 J **(d)** 2 J

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define work in the scientific sense. Give its SI unit. 2

12. State the work-energy theorem. 2

13. What is the use of a fixed pulley? 2

14. When is the work done negative? Give an example. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A girl lifts a dumbbell of 5 kg to a height of 1.2 m and then lowers it. State when positive and when negative work is done and calculate the work done by her force in lifting (g = 10). 3

16. Calculate the energy consumed and the cost if a 100 W bulb is used for 10 hours and the rate is 6 rupees per unit. 3

17. A boy of mass 40 kg climbs 12 stairs of 20 cm each in 8 s. Find his power (g = 10). 3

OR

Derive the expression for the kinetic energy of a body of mass m moving with velocity v. 3

18. A person raises a load of 30 cm height using a 40 cm wide ramp. Find the length of the ramp and the mechanical advantage. 3

OR

A force of 20 N moves a block through 3 m while friction of 8 N opposes the motion. Find the work done by the force, the work done by friction and the net work. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the work done by a constant force, with the cases of force along, opposite and perpendicular to the displacement. Calculate the work done by a force of 15 N moving a body 4 m along the force. 5

OR

Explain the conservation of energy in a swinging pendulum. Describe the changes of KE and PE and the effect of air resistance. 5

20. Derive the work-energy theorem and use it to find the speed of a 2 kg body at rest after a net work of 100 J is done. 5

OR

A pump lifts 1000 litres of water per minute from a well 30 m deep. Find the power required if the efficiency is 100 per cent and the power input if the efficiency is 75 per cent (1 litre = 1 kg, g = 10). 5

END OF QUESTION PAPER Total: 40 Marks